

Maven POM.xml – Complete Interview Notes

1) What is POM.xml?

POM stands for Project Object Model. It is the core configuration file in Maven where we define project details, dependencies, plugins, and build configuration.

2) Dependency in POM.xml

Dependencies are external libraries required for the project. We define them in pom.xml so Maven downloads them automatically.

```
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-core</artifactId>
  <version>5.3.0</version>
</dependency>
```

3) Packaging Concept

When we build the project, Maven creates a package like JAR or WAR. The package name is usually:

`artifactId-version.jar`

Example: `myapp-1.0.jar`

4) Why Unique Naming?

Project name must be globally unique. So we use reverse domain name to avoid conflicts.

5) GroupId Explained

GroupId represents the organization or company name.

It is written in reverse domain format.

Example:
`com.company.project`

Purpose: • Uniquely identifies organization • Avoids conflicts • Groups related projects

6) ArtifactId Explained

ArtifactId is the name of the project or module.

Example:
student-management-system

Purpose: • Identifies project • Used in generated JAR/WAR name

7) Version Explained

Version represents the current version of the project.

Examples:
1.0
1.0-SNAPSHOT

• SNAPSHOT → Development version • RELEASE → Stable version

8) Complete Example

```
<project>
  <groupId>com.company</groupId>
  <artifactId>myapp</artifactId>
  <version>1.0</version>
</project>
```

9) After Adding Dependency in Eclipse

Steps to refresh Maven project in Eclipse:

1. Right click project
2. Select Maven → Update Project
3. Click OK

Shortcut: Alt + F5

10) Interview Summary

POM.xml is the heart of Maven project. It defines dependencies, project identity (groupId, artifactId, version), and build configuration. Maven uses it to manage and build the project efficiently.

11) Quick Interview Q&A;

Question	Answer
What is POM?	Project Object Model

GroupId?	Organization name
ArtifactId?	Project name
Version?	Project version
Why reverse domain?	To ensure uniqueness